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The Editor
Economic Analysis and Policy

Dear Editor,

I am pleased to submit for your consideration the enclosed manuscript, **"The Solow Residual as a Measurement Artefact: Investment Gestation Lags, Intangible Capital, and Macroeconomic Policy"**, for publication in *Economic Analysis and Policy*.

The paper argues that a measurable share of what is labelled total factor productivity is neither technology nor innovation, but a bookkeeping artefact produced by two conventions standard growth accounting imposes at zero: the gestation lag between investment and output is treated as constant and negligible, and produced capital omits intangible investment. Using OECD and middle-income data, correcting for these two factors reconciles production-side and wealth-side national accounts to within 1–2 % for most countries, raises measured TFP by about 1.7 percentage points, raises the implied labour share by 1.7 percentage points, and raises produced capital by up to 1.1 % (total wealth by up to 0.3 %) in research-intensive economies. The results are largest where investment has shifted toward long-gestation intangible assets. These are measurement consequences with direct implications for fiscal-policy rules, monetary-policy transmission, and cross-country GDP comparability, rather than forecasting exercises.

I believe this work is well-suited to *Economic Analysis and Policy* for three reasons, and I would suggest submission under either the journal's "Modelling Economic Policy Issues" or "Analyses of Topical Policy Issues" category.

First, the journal's stated focus is on research with strong policy relevance, and the paper's central finding is precisely that: standard TFP measures contain a structural tempo bias that misdirects potential-output estimates, output-gap calculations, and the calibration of fiscal and monetary policy responses. Removing the bias reallocates a median 1.7 percentage points of log-level TFP and changes the implied labour share by a comparable amount.

Second, the analysis provides new empirical tools for applied macroeconomic policy. The "Relational PIM" diagnostic adapts Brass relational methods to capital accounting, and the demographic tempo-and-quantum correspondence is used to construct an observable, parameter-free proxy for the time-varying investment-to-output lag. Both tools are designed to be reused across countries, vintages, and asset classifications.

Third, the corrections are disciplined by a wealth-accounting identity, not only by production-side residuals. Reconciling flow-side and stock-side capital accounts to within 1–2 % after correcting the lag and intangible share provides an external validity check, and the Supporting Information reports an expanded set of robustness checks: Monte Carlo identification sharpness calibrated to four countries (United States, Republic of Korea, France and Colombia), a leave-one-out robustness check for historical episodes, a cross-country γ_{price} sensitivity summary, a 1.5×–3.0× R&D scaling sensitivity for intangible capital, and cluster-validation metrics. The manuscript also now opens with a short, non-technical policy hook that explains why the measurement issue matters for fiscal and monetary policy.

I confirm the following:

- This work is original and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All data sources (Penn World Table 10.01; World Bank CWON; World Bank WDI; OECD GFCF) are publicly available; frozen extracts, code, and intermediate outputs are archived in the companion repository at https://github.com/bougtoir/gdp-tempo-paper.
- I am the sole author.
- I have no competing interests, financial or otherwise, related to this work.
- I used generative AI to assist with formatting the text, choosing words, and writing analysis code; this is documented in the Declarations section of the title page.

A title page with full author details and a Declarations section is included as a separate file for the double-anonymized review process.

I would be grateful for your consideration.

Yours sincerely,

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